Todo App – Docker Assignment

Express (Node.js) frontend + Flask backend, containerised with Docker

1. Frontend (Node.js with Express)

- Created the frontend using Express and Node.js (**frontend/server.js**, **package.json**, **public/**).
- The form is configured to send its request to the Flask backend.



Figure 1: frontend/ folder with server.js, package.json and public/

2. Backend (Flask)

- The Flask backend (**backend/app.py**) handles the form submission and processes the data.
- Dependencies are listed in **requirements.txt**.

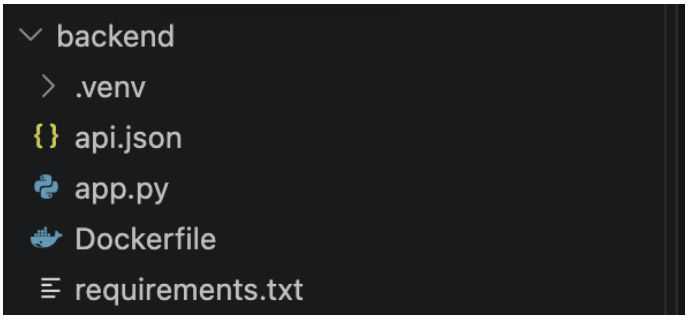


Figure 2: backend/ folder with app.py, requirements.txt and Dockerfile

3. Folder Structure

- The project is organised with separate **frontend** and **backend** folders, plus **compose.yaml**, **.dockerignore**, **.gitignore** and **README.md** at the root.

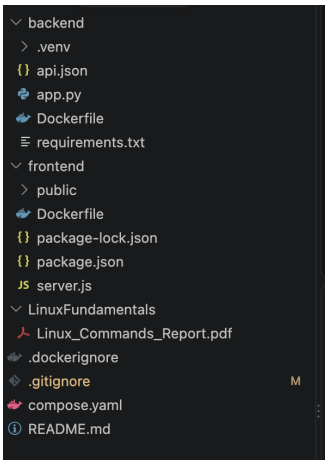


Figure 3: Project folder structure in VS Code

4. Docker Configuration

- A Dockerfile was created for both the frontend and the backend.
- **compose.yaml** (Docker Compose) connects both services on the same network.
- Both images were uploaded to Docker Hub; code pushed to GitHub with **node_modules**, **.vscode** etc. in **.gitignore**.

Building the frontend image

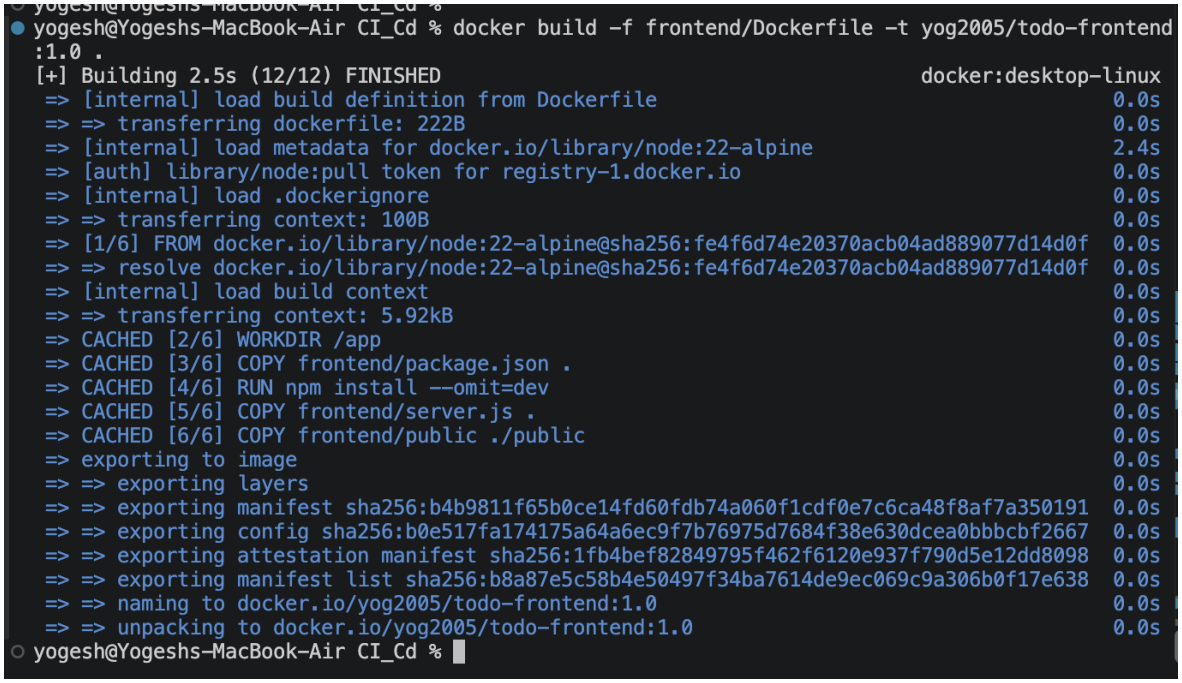


Figure 4: docker build -f frontend/Dockerfile -t yog2005/todo-frontend:1.0 .

Logging in to Docker Hub

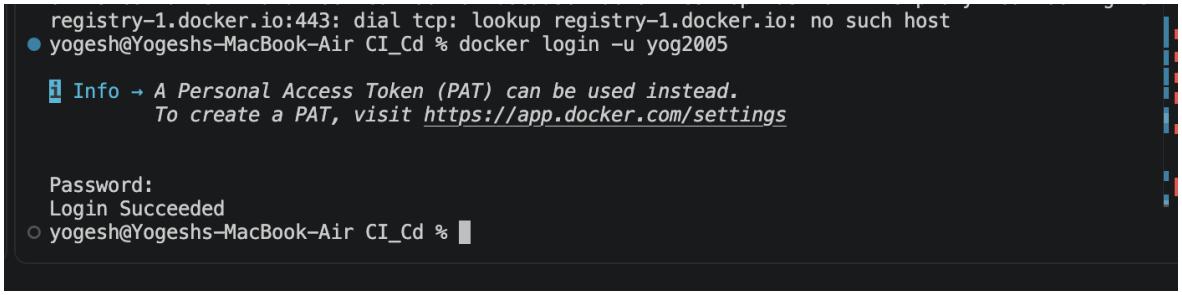


Figure 5: docker login -u yog2005 – Login Succeeded

Pushing the backend image



Figure 6: Backend image pushed to docker.io/yog2005/todo-backend (tag 1.0)

Pushing the frontend image

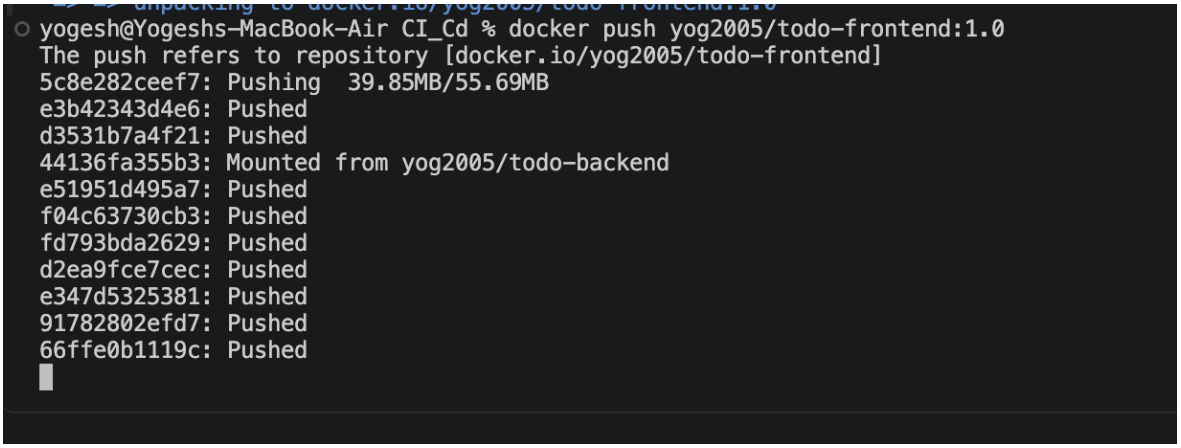


Figure 7: docker push yog2005/todo-frontend:1.0